

RaceXL – Excel Score Sheet Generator

User Instructions

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ניתן למצוא גרסה של הוראות שימוש אלו בשפה העברית בכתובת:

<https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Instructions-heb.pdf>

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1. Introduction

“RaceXL - Excel Score Sheet Generator” is a utility application designed to automate and streamline the creation of structured Excel score sheets for the “Race to the Technion” event. The app generates Excel workbooks with standardized scoring templates from a JSON configuration file.

It enables fast score calculation, real-time tracking, and a significant reduction in the manual workload and errors involved in event setup.



The app's icon

Originally, the only intended use for the app was to read the JSON configuration file and then output an Excel spreadsheet named “race_to_the_technion.xlsx”, which would then be shared with the stations’ managers via Google Sheets, for them to input each team's score as they go. This would enable real-time scoring updates and save even more time. All additional options were added later for convenience.

A full technical documentation of the app can be found at:

<https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Documentation-en.pdf>

Important Note

An agreement was reached with the Odyssey Program students.

The students agreed to add to the race a **permanent** question about me, Amit Zlatkin, the creator of the app, similar to the permanent question about the name of the race’s original inventor.

In exchange, I have released the app as open-source, and all information and code files are openly available in the app’s GitHub repository:

<https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker>

Known Issues

1. The supported Hebrew input is poorly printed in the terminal on Windows. It may be fixed in the future, depending on demand.
2. The issue with Microsoft Defender, mentioned in section 3 later.

2. Installation

Getting started is straightforward. No complex installers are required.

To install the app, download the executable for your specific platform (Windows or Linux) from the latest release, here:

<https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/releases>

Make sure to download the correct file.

3. Launching the App

RaceXL is a **command-line tool** that relies on Unix-style arguments and flags.

Windows

- Double-click executable (recommended)

For the standard, cleanest execution, simply double-click the application file, and there is no need to type the `./RaceXL` command at the start.

- Terminal

You can also use Command Prompt (cmd) or PowerShell. If you decide to run the app this way, ensure you use the backslash (\) and include the `.exe` extension, if necessary (e.g., `.\RaceXL.exe`).

A note about Windows SmartScreen & Microsoft Defender

Because *RaceXL* is an independent utility, Windows SmartScreen or Microsoft Defender may warn you when you try to launch it for the first time.

Click **"More info"** → **"Run anyway"** to launch the app.

This happens because the app isn't widely downloaded or recognized by Microsoft's cloud-based reputation system. Said recognition will start only at a few hundred downloads.

Also, the app is missing a "digital signature". It costs hundreds of dollars per year, so I didn't even buy this "trusted certificate", let alone sign the code with it.

Linux (Ubuntu)

- Terminal

Open your terminal and execute the application using the standard forward-slash command syntax: `./RaceXL`.

4. Command Syntax & Flags

The general, Unix-style command syntax is:

```
./RaceXL [help] [name] [configuration]
```

The app supports the following flags to modify its behavior:

Flag	Long Version	Description
-h	--help	Displays the help menu and usage instructions. Must be the first flag used.
-n	--name	Sets a custom name for the output Excel file.
-d	--default	Skips user prompt and uses the built-in default configuration.

The long and short versions of the flags are interchangeable.

All Execution Scenarios

RaceXL adapts to your specific needs through seven distinct execution patterns:

Guided Default

```
» ./RaceXL
```

Use the default filename and give a prompt to confirm the default configuration.

Help Menu

```
» ./RaceXL --help
```

Shows the help message.

Silent Default

```
» ./RaceXL --default
```

Generates a file with the default name and settings without questions.

Targeted Configuration

```
» ./RaceXL race-config.json
```

The original use case – keeps the default filename but uses your specific JSON data.

Named Output

```
» ./RaceXL --name race-scores.xlsx
```

Sets a custom file name while still being prompted for configuration.

Full Control

```
» ./RaceXL --name race-scores.xlsx race-config.json
```

Defines both the output name and the data source manually.

Custom Name & Silent Run

```
» ./RaceXL --name race-scores.xlsx --default
```

Uses a custom name with the built-in default configuration.

5. Tips for Path and Space Handling

- Relative Paths

You can create the Excel file in a different folder and reference JSON files in other folders using relative directory syntax. For example:

```
./RaceXL --name ./races/race.xlsx ./configs/config.json
```

Also, you can use forward slashes for the paths everywhere! Even on Windows!

- Handling Spaces

If your file name or path has spaces, enclose it in single (' ') or double (" ") quotes. For example:

```
./RaceXL --name "2026 Race.xlsx" '2026 Config.json'
```

- Automatic Extensions

If you omit or forget to type the .xlsx or .json extensions, the app will automatically add the correct extension for you.

Together, these tips allow us to use the following clean command:

```
./RaceXL --name "./.../2026 Race" './.../2026 Config'
```

Where each “...” stands for some path.

6. JSON Configuration File Format

The system uses a JSON file to define the structure of the Excel sheet.

- Root Fields:

- **teams** (integer): Number of participating teams.
- **activities** (array): List of activity objects.

- Activity Object Fields:

- **name** (string): Activity name, *can be in Hebrew*.
- **location** (string): Activity location, *can be in Hebrew*.
- **makeWorksheet** (boolean): Create an activity sheet or not (Human Pyramid doesn't need a worksheet). The following fields are relevant only when this field is true.
- **rows** (integer): Number of scoring rows.
- **numberTable** (boolean): Activity results input type.
 - true → numeric input
 - false → checkbox input
- **pointsPerRow** (integer): Points per “completed row”, i.e., per selected checkbox. This field is relevant only when **numberTable** is false.

Note: It is best and safest to specify values for irrelevant fields as well.

- Example Layout & JSON:

The app's GitHub repository contains an example of a race layout and the JSON file that is associated with it at the following links:

- Example race layout:
 - In English: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Layout-en.pdf>
 - In Hebrew: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Layout-heb.pdf>
- Associated JSON files:
 - In English: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Config-en.json>
 - In Hebrew: <https://github.com/AmitZlatkin/Race-to-The-Technion-Excel-Maker/blob/main/Instructions/Example-Race-Config-heb.json>

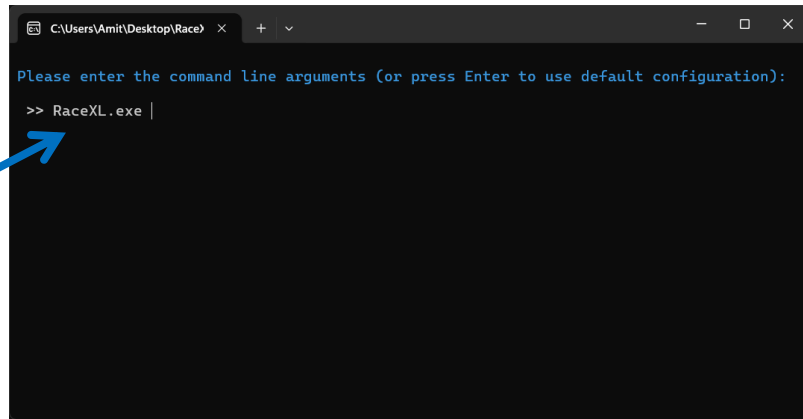
7. Screenshots

The screenshots were taken within the double-click icon app for Windows.

The view in the terminals is very similar.

The window that opens:

Notice how the program name is already there – no need to type it!

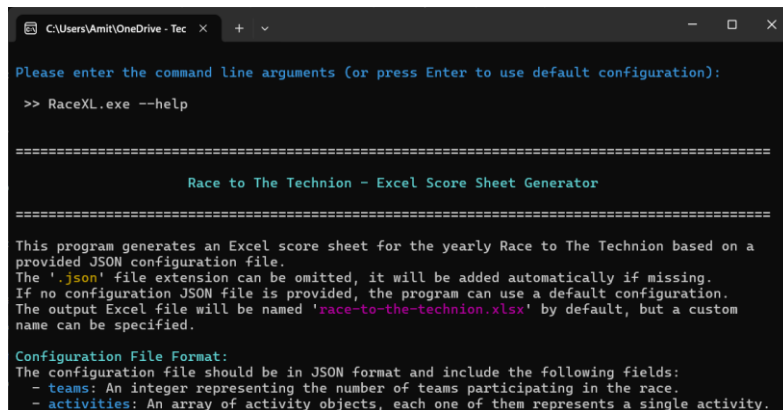


```
C:\Users\Amit\Desktop\Race> x + v - □ ×

Please enter the command line arguments (or press Enter to use default configuration):

>> RaceXL.exe |
```

The start of the help message:



```
C:\Users\Amit\OneDrive - Tec> x + v - □ ×

Please enter the command line arguments (or press Enter to use default configuration):

>> RaceXL.exe --help

=====

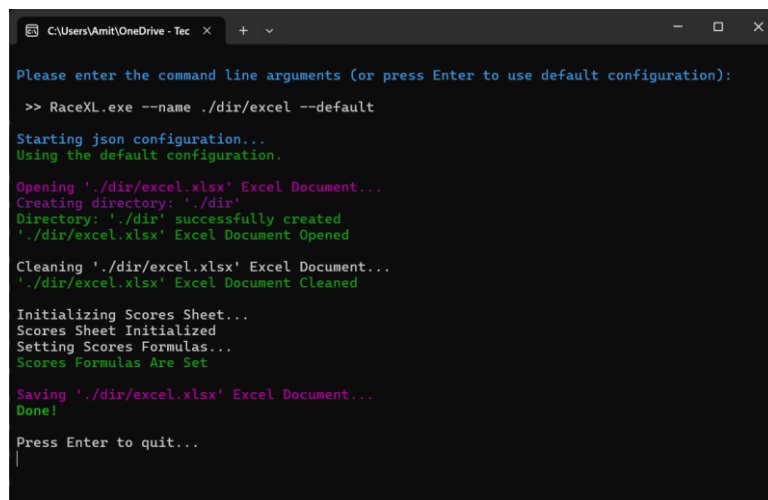
Race to The Technion - Excel Score Sheet Generator

=====

This program generates an Excel score sheet for the yearly Race to The Technion based on a provided JSON configuration file.
The '.json' file extension can be omitted, it will be added automatically if missing.
If no configuration JSON file is provided, the program can use a default configuration.
The output Excel file will be named 'race-to-the-technion.xlsx' by default, but a custom name can be specified.

Configuration File Format:
The configuration file should be in JSON format and include the following fields:
- teams: An integer representing the number of teams participating in the race.
- activities: An array of activity objects, each one of them represents a single activity.
```

The output of the command: `./RaceXL --name ./dir/excel --default`



```
C:\Users\Amit\OneDrive - Tec> x + v - □ ×

Please enter the command line arguments (or press Enter to use default configuration):

>> RaceXL.exe --name ./dir/excel --default

Starting json configuration...
Using the default configuration.

Opening './dir/excel.xlsx' Excel Document...
Creating directory: './dir'
Directory: './dir' successfully created
'./dir/excel.xlsx' Excel Document Opened

Cleaning './dir/excel.xlsx' Excel Document...
'./dir/excel.xlsx' Excel Document Cleaned

Initializing Scores Sheet...
Scores Sheet Initialized
Setting Scores Formulas...
Scores Formulas Are Set

Saving './dir/excel.xlsx' Excel Document...
Done!

Press Enter to quit...
|
```


The resulting Excel file:

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K
1	תחנה	מיקום	קבוצה 1	קבוצה 2	קבוצה 3		זמנים	קבוצה 1	קבוצה 2	קבוצה 3	
2	Activity A	Location A					זמן התחלה				
3	Activity B	Location B					זמן סיום				
4	Activity C	Location C					הפרש	0	0	0	
5	סך הכל		0	0	0						
6											
7											
8											
9											